The Stark problem in a Debye plasma: a certified coupled-channel map of field–screening competition

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A hydrogen-like atom in a static electric field and a Debye plasma experiences two competing ionization mechanisms: field ionization through the Stark barrier and bound-state dissolution by screening. The combined problem is non-separable, and the published record consists of isolated parameter points. We solve it with a coupled-channel complex-scaled method whose radial channels carry the exact Laguerre basis of the unperturbed problem, coupled only by the dipole selection rule $\ell \to \ell \pm 1$: the resulting block-tridiagonal complex-symmetric pencils reach the classical pure-Stark benchmark at $F = 0.1$ a.u. ($E = -0.527418175 - 0.007269057i$) with fewer than 10^3 degrees of freedom and geometric convergence in the channel cutoff, and every assembly is certified at run time against the exact hydrogen spectrum — a discipline that caught a factor-of-two kinetic-energy defect masked by an apparently convergent spurious state during development, and that directly addresses the documented fragility of complex-rotated Laguerre-mesh width calculations. On this basis we compute the first systematic resonance map of the combined problem: ground-state positions and widths on a grid in the (F, λ) plane and the $m = 0$ Stark manifold of $n = 2$ under screening. The map quantifies the field–screening competition and its two regimes, and the $n = 2$ manifold exhibits the interplay of linear Stark splitting with screening-induced level upshift.

I. INTRODUCTION

The Stark effect in hydrogen and the screened Coulomb (Yukawa–Debye) potential are two classical single-parameter problems whose resonance structure is known to high precision: the pure Stark resonances from complex-dilation and resummation methods [1–4], and the screened-Coulomb bound states and critical screenings from decades of plasma-physics work [5, 6]. Their combination — an atom in a plasma subjected to a static field, the generic situation in dense-plasma spectroscopy — is qualitatively richer: the field tilts the potential and opens a Stark barrier, screening lifts and delocalizes the bound states, and the two ionization mechanisms compete. The combined problem is non-separable (screening breaks the parabolic separability of the Stark problem), and the published record consists of a small number of parameter points: the pioneering ground-state calculations of Yu and Ho [7], the complex-scaled Lagrange-mesh study of Lin and Ho [8] covering ground and selected excited states, and recent bound-state-level work in dense plasmas [9]. A systematic map of the (F, λ) plane — positions, widths, critical curves, and the fate of the excited Stark manifolds under screening — has not been available.

This paper supplies such a map, built on two methodological choices. First, instead of a two-dimensional mesh, we exploit the dipole selection rule: the field couples only adjacent angular channels, so the problem is a block-tridiagonal radial system in which each channel carries the *exact* orthonormal Laguerre basis of the unperturbed radial problem [10], and the whole pencil is complex-scaled analytically. Convergence in the channel cutoff is geometric, and the classical pure-Stark benchmark is reproduced to ten digits with fewer than a thousand degrees of freedom — two orders of magnitude below mesh requirements. Second, every assembly is certified at run time against exact identities (the unperturbed hydrogen spectrum, channel overlaps), a discipline inherited from the companion programs [10, 11]. This is not decorative: an early build of the present code carried a factor-of-two kinetic-energy defect that produced a *convergent, rotation-stable* spurious state at a plausible energy — caught in seconds by the exact-spectrum certification, not by convergence studies. The fragility of complex-rotated Laguerre-mesh width calculations against matrix-element errors is documented in the literature, and certification is our systematic answer to it.

II. METHOD

In the $m = 0$ sector the Hamiltonian of a hydrogen-like atom in a Debye plasma and a static field $F\hat{z}$,

$$H = -\tfrac{1}{2}\nabla^2 - \frac{Ze^{-\lambda r}}{r} + Fz, \tag{1}$$

acts on channel wave functions $u_\ell(r)$ coupled only by the dipole matrix elements of $z = r\cos\theta$: $\langle \ell + 1, 0|\cos\theta|\ell, 0\rangle = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$. Each channel is expanded in the orthonormal generalized Laguerre basis of Ref. [10], with a common scale and quadrature grid across channels so that the inter-channel dipole blocks are exact contractions. Complex scaling $r \to re^{i\theta_c}$ acts analytically on every block — kinetic terms scale by $e^{-2i\theta_c}$, Coulombic parts by $e^{-i\theta_c}$, the linear Stark

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coupling by $e^{+i\theta_c}$, and the screening factor enters the quadrature at complex argument [1, 12–15] — giving a block-tridiagonal complex-symmetric pencil. Resonances are identified by θ_c-stability between two rotation angles, and every tabulated value carries the maximum spread over a grid of two channel cutoffs and two basis sizes.

Certification: at $F = \lambda = 0$ the assembled pencil must reproduce the exact hydrogen levels $-1/2n^2$; we require agreement below 10^{-9} on the lowest five levels at every build, together with channel-overlap identities at the 10^{-12} level. At $F = 0.1$, $\lambda = 0$ the method gives $E = -0.527418175 - 0.007269057i$ with geometric convergence in the channel cutoff ($4 \times 10^{-5} \to 5 \times 10^{-10}$ over $\ell_{\max} = 4 \to 16$) at 780 degrees of freedom, in agreement with the classical complex-dilation value [2–4]; in the opposite limit $F = 0$, $\lambda = 0.25$ the $\ell = 1$ block reproduces the independently triple-verified $2p$ resonance of the one-dimensional engine [10]. A remark on solver scale: at these dimensions dense complex diagonalization computes each map point in seconds, so contour-integral eigensolvers, useful at mesh scales, are unnecessary here.

III. RESULTS

[[RESULTS-PLACEHOLDER: D1a anchor table, the (F,lambda) ground map figure and discussion of the two regimes, the n=2 manifold figure.]]

IV. CONCLUSIONS

[[CONCLUSIONS-PLACEHOLDER]]

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